

```
In [1]: # !pip install langchain
```

```
In [2]: # !pip install openai
```

```
In [3]: # !pip install wikipedia
```

```
In [4]: # !pip install numexpr
```

```
In [5]: import os
from dotenv import load_dotenv
load_dotenv()
OPENAI_API_KEY = os.getenv("OPENAI_API_KEY")
```

```
In [6]: from langchain_classic.chat_models import ChatOpenAI
llm = ChatOpenAI(temperature=0, # 창의성 0으로 설정
                model_name='gpt-4.1-mini', # 모델명
                )

from langchain_classic.agents import load_tools
from langchain_classic.agents import initialize_agent
from langchain_classic.agents import AgentType

tools = load_tools(["wikipedia", "llm-math"], llm=llm) #llm-math의 경우 나이 계산을 위해 사용
agent = initialize_agent(tools,
                        llm,
                        agent=AgentType.ZERO_SHOT_REACT_DESCRIPTION,
                        description='계산이 필요할 때 사용',
                        verbose=True)

agent.run("에드 시런이 태어난 해는? 2026년도 현재 에드 시런은 몇 살?")
```

```
C:\Users\User\AppData\Local\Temp\ipykernel_21468\437185133.py:2: LangChainDeprecationWarning: The class `ChatOpenAI` was deprecated in LangChain 0.0.10 and will be removed in 1.0. An updated version of the class exists in the `langchain-openai` package and should be used instead. To use it run `pip install -U `langchain-openai` and import as `from `langchain_openai import ChatOpenAI``.
```

```
llm = ChatOpenAI(temperature=0, # 창의성 0으로 설정
```

```
C:\ProgramData\anaconda3\Lib\site-packages\keras\src\export\tf2onnx_lib.py:8: FutureWarning: In the future `np.object` will be defined as the corresponding NumPy scalar.
```

```
if not hasattr(np, "object"):
```

```
WARNING:tensorflow:From C:\ProgramData\anaconda3\Lib\site-packages\tf_keras\src\losses.py:2976: The name tf.losses.sparse_softmax_cross_entropy is deprecated. Please use tf.compat.v1.losses.sparse_softmax_cross_entropy instead.
```

```
C:\Users\User\AppData\Local\Temp\ipykernel_21468\437185133.py:11: LangChainDeprecationWarning: LangChain agents will continue to be supported, but it is recommended for new use cases to be built with LangGraph. LangGraph offers a more flexible and full-featured framework for building agents, including support for tool-calling, persistence of state, and human-in-the-loop workflows. For details, refer to the [LangGraph documentation](https://langchain-ai.github.io/langgraph/) as well as guides for [Migrating from AgentExecutor](https://python.langchain.com/docs/how_to/migrate_agent/) and LangGraph's [Pre-built ReAct agent](https://langchain-ai.github.io/langgraph/how-to/create-react-agent/).
```

```
agent = initialize_agent(tools,
```

```
C:\Users\User\AppData\Local\Temp\ipykernel_21468\437185133.py:17: LangChainDeprecationWarning: The method `Chain.run` was deprecated in langchain-classic 0.1.0 and will be removed in 1.0. Use `invoke` instead.
```

```
agent.run("에드 시런이 태어난 해는? 2026년도 현재 에드 시런은 몇 살?")
```

> Entering new AgentExecutor chain...

Question: 에드 시런이 태어난 해는? 2026년도 현재 에드 시런은 몇 살?

Thought: 먼저 에드 시런이 태어난 해를 찾아야 한다. 그 후 2026년에서 태어난 해를 빼서 나이를 계산한다.

Action: wikipedia

Action Input: Ed Sheeran birth year

Observation: Page: Ed Sheeran

Summary: Edward Christopher Sheeran (SHEER-ən; born 17 February 1991) is an English singer-songwriter and musician. Born in Halifax, West Yorkshire, and raised in Framlingham, Suffolk, he began writing songs around the age of eleven. In early 2011, Sheeran independently released the extended play No. 5 Collaborations Project. He signed with Asylum Records the same year.

Sheeran's debut album, + ("Plus"), was released in September 2011 and topped the UK Albums Chart. It contained his first hit single, "The A Team". In 2012, Sheeran won the Brit Awards for Best British Male Solo Artist and British Breakthrough Act. Sheeran's second studio album, × ("Multiply"), topped charts around the world upon its release in June 2014. It was named the second-best-selling album worldwide of 2015. In the same year, × won Album of the Year at the 2015 Brit Awards, and he received the Ivor Novello Award for Songwriter of the Year from the British Academy of Songwriters, Composers and Authors. A single from ×, "Thinking Out Loud", earned him the 2016 Grammy Awards for Song of the Year and Best Pop Solo Performance.

Sheeran's third album, ÷ ("Divide"), was released in March 2017 and was the best-selling album worldwide of 2017. The first two singles from the album, "Shape of You" and "Castle on the Hill", broke records in several countries by debuting in the top two positions on the charts. He also became the first artist to have two songs debut in the US top 10 in the same week. By March 2017, Sheeran had accumulated ten top-10 singles from ÷ on the UK Singles Chart, breaking the record for most top-10 UK singles from one album. His fourth single from ÷, "Perfect", reached number one in the US, Australia, and the UK, where it became the Christmas number one in 2017. The world's best-selling artist of 2017, he was named the Global Recording Artist of the Year. Released in 2019, his fourth overall and first collaborative album, No.6 Collaborations Project, debuted at number one in most major markets, and spawned three UK number one singles, "I Don't Care", "Beautiful People", and "Take Me Back to London". His fifth studio album, = ("Equals"), topped the charts in most major markets in 2021. His sixth album, - ("Subtract"), was released on 5 May 2023, while his seventh album, Autumn Variations, was released on 29 September 2023 under his own record label, Gingerbread Man Records. Sheeran's eighth studio album, Play was released on 12 September 2025.

Sheeran has sold 200 million records worldwide, making him one of the world's best-selling music artists. He has 119 million RIAA-certified units in the US, and two of his albums are in the list of the best-selling albums in UK chart history. In December 2019, the Official Charts Company named him its artist of the decade, with the most combined success in the UK album and singles charts in the 2010s. As of July 2024, he is the third most followed artist on Spotify. Several of his tours are amongst the most-attended concert tours, and his ÷ Tour became the highest-grossing of all time in August 2019. An alumnus of the National Youth Theatre in London, Sheeran's acting roles include appearing in the 2019 film Yesterday, playing himself. In 2025, Time magazine listed him as one of the world's 100 most influential people.

Page: = (album)

Summary: = (Equals) is the fifth studio album by English singer-songwriter Ed Sheeran. It was released on 29 October 2021 through Asylum and Atlantic Records. The album was supported by five singles: "Bad Habits", "Shivers", "Overpass Graffiti", "The Joker and the Queen", and "2step". The song "Visiting Hours" was released as a promotional single.

Upon release, = received generally mixed reviews from music critics, who appreciated Sheeran's emerging sonic experimentation while criticising its production. The album reached number one in Australia, Belgium, Canada, Denmark, France, Germany, Ireland,

Italy, Lithuania, the Netherlands, New Zealand, Scotland

Thought: 에드 시런은 1991년 2월 17일에 태어났다. 2026년 현재 그의 나이는 $2026 - 1991 = 35$ 세이다.

Final Answer: 에드 시런은 1991년에 태어났으며, 2026년 현재 35세입니다.

> Finished chain.

Out[6]: '에드 시런은 1991년에 태어났으며, 2026년 현재 35세입니다.'

In []: